torch.linalg.slogdet#

https://pytorch.org/docs/stable/generated/torch.linalg.slogdet.html#torch.li

Description:

Computes the sign and natural logarithm of the absolute value of the determinant of a square matrix. For complex A, it returns the sign and the natural logarithm of the modulus of the determinant, that is, a logarithmic polar decomposition of the determinant. The determinant can be recovered as sign * exp(logabsdet). When a matrix has a determinant of zero, it returns (0, -inf). Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

Function Signature

```
torch.linalg.slogdet(A, *, out=None)#
Parameters
Keyword Arguments
Returns
```

Parameters

Keyword

out (tuple, optional) – output tuple of two tensors. Ignored if None. Default: None.

Returns:

A named tuple (sign, logabsdet). sign will have the same dtype as A. logabsdet will always be real-valued, even when A is complex.

Code Examples

```
>>> A = torch.randn(3, 3)
>>> A
tensor([[ 0.0032, -0.2239, -1.1219],
        [-0.6690,  0.1161,  0.4053],
        [-1.6218, -0.9273, -0.0082]])
>>> torch.linalg.det(A)
tensor(-0.7576)
>>> torch.logdet(A)
tensor(nan)
>>> torch.linalg.slogdet(A)
torch.return_types.linalg_slogdet(sign=tensor(-1.),
logabsdet=tensor(-0.2776))
```

Notes & Warnings

NOTE See also torch.linalg.det() computes the determinant of square matrices.

Detailed Documentation

Documentation

Computes the sign and natural logarithm of the absolute value of the determinant of a square matrix.

For complex A, it returns the sign and the natural logarithm of the modulus of the determinant, that is, a logarithmic polar decomposition of the determinant.

The determinant can be recovered as sign * exp(logabsdet). When a matrix has a determinant of zero, it returns (0, -inf).

Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

torch.linalg.det() computes the determinant of square matrices.

A (Tensor) – tensor of shape (*, n, n) where * is zero or more batch dimensions.

out (tuple, optional) – output tuple of two tensors. Ignored if None. Default: None.

A named tuple (sign, logabsdet).

sign will have the same dtype as A.

logabsdet will always be real-valued, even when A is complex.